Universal Closure Regimes: A Geometric Taxonomy of Architectural and Additive Manifolds

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1 Abstract

We formalize a structural transition from the angular manifold routing frameworks established in Allard [1] toward a generalized theory of geometric closure (σ). This investigation defines the regression toward an isotropic zero-point attractor ($\sigma \to 1$) as a universal classifier for both sacred architectural archetypes and the additive gap-structures of prime k-tuples. By grounding our measurements in Fisher-information-grounded closure functionals and mapping these states onto the Poincaré upper half-plane ($\mathbb{H}$), we identify a distinctive e^2 regime ladder as the primary structural determinant for morphological stability. We demonstrate that the tangent operator $\mathcal{T}(X)$, functioning as a projective slope, is mathematically equivalent to the hyperbolic arclength Λ along an isomorphic geodesic trajectory, providing a unified metric for Assessing structural survival across disparate information-geometric manifolds.

2 Introduction: From Angular Non-Uniformity to Isotropic Regression

The phenomenon of angular non-uniformity in L2-normalized token embeddings, recently exploited for sublinear compute reduction in transformer architectures, suggests the existence of a broader geometric law. As detailed in the previous preprint "Angular Manifold Routing: Sublinear Compute Reduction via Hopf-Base Sector Discretization" [1], fixed geometric routing via Hopf fibration provides significant optimization in expert path efficiency. However, we posit that this non-uniformity is a specific manifestation of a hidden "4th axis"—a dimension of geometric reconciliation.

While the primary axes x, y, z quantify spatial extent, this 4th axis, defined by the closure metric σ, represents the informational entropy of the form's distribution. The strategic objective is to identify the hidden dimension of geometric reconciliation that allows high-dimensional state spaces to collapse into stable, low-dimensional attractors. By transitioning from the computational utility of Hopf-base discretization to the fundamental information-geometric metric of σ, we define a universal coordinate of isotropic regression.

3 Mathematical Framework: Metrics of Closure and Symmetry

To classify the departure from isotropy in complex state spaces, we define a raw state as a positive n-tuple $X = (x_1, \ldots, x_n)$, normalized to a probability simplex $\mathcal{S}_n$ where $p_i = x_i / \sum x_j$.

3.1 The Fisher Information Metric

As Information Geometers, we adopt the Fisher Information Metric as the natural Riemannian metric on the probability simplex:

$$ds^2 = \sum_{i=1}^{n} \frac{dp_i^2}{p_i}$$

This metric justifies the use of KL-divergence as the fundamental measure of distance from the uniform distribution $u_i = 1/n$.

3.2 Defining the Closure Score (σ)

We utilize two primary quadratic and entropic functionals to measure structural closure:

- **Quadratic Closure** (σ_Q):

$$\sigma_Q(X) = 1 - \frac{\sum_{i=1}^{n}(p_i - \frac{1}{n})^2}{1 - \frac{1}{n}}$$

- **Entropic/KL Closure** (σ_{KL}):

$$\sigma_{\mathrm{KL}}(X) = 1 - \frac{D_{\mathrm{KL}}(p\|u)}{\log n} = 1 - \frac{\sum_i p_i \log(np_i)}{\log n}$$

3.3 The Symmetry Invariant (τ)

Closure alone cannot distinguish between forms with identical σ but divergent structural intents (e.g., a pyramid vs. a processional ark). We introduce the symmetry-class invariant $\tau(X)$ to map axial differentials:

$$\Delta_h = |x - y|, \quad \Delta_v = |x - z| + |y - z|$$

$$\tau(X) = \frac{\Delta_v - 2\Delta_h}{\Delta_v + 2\Delta_h}$$

3.4 Symmetry Fingerprinting

The full state vector $(X, \sigma, \mathbf{T})$ incorporates the fingerprint $\mathbf{T}(X) = (T_{\text{cube}}, T_{\text{ascent}}, T_{\text{carrier}})$, defining the morphological family:

Family Component	Definition	Morphological Indication
T_{cube}	$\sigma(X)(1-\tau(X))$	Planar stability; vertical differentiation.
T_{ascent}	$\sigma(X)\max(\tau(X),0)$	High vertical ascent features.
T_{carrier}	$\sigma(X)\max(-3\tau(X),0)$	Directional/processional bias; horizontal elongation.

4 Complex Manifold Embedding and Hyperbolic Integration

Identifying boundary-crossing dynamics necessitates mapping Euclidean measurements into complex and Riemannian manifolds to analyze the trajectory toward the isotropic origin.

4.1 Complex Semicircular Mapping

The Complex Bridge embedding $\mathcal{Z}(X)$ represents a half-turn complex orbit from minimal closure to the zero-point:

$$\mathcal{Z}(X) = i(e^{i\pi\sigma(X)} + 1)$$

This maps $\sigma = 0$ (maximal anisotropy) to $2i$ and $\sigma = 1$ (full closure) to 0.

4.2 The Tangent Operator ($\mathcal{T}$) and Hyperbolic Arclength (Λ)

To linearize this circular flow, we apply a projective reduction using the tangent half-angle substitution (Weierstrass substitution). The Tangent Operator $\mathcal{T}(X)$ is defined as:

$$\mathcal{T}(X) = \tan\left(\frac{\pi\sigma(X)}{2}\right)$$

When the bridge is embedded into the Poincaré upper half-plane ($\mathbb{H}$), the operator $\mathcal{T}(X)$ is mathematically equivalent to the hyperbolic arclength Λ along the vertical geodesic descending from the anchor $(0,2)$ toward the ideal point 0:

$$\Lambda(X) = -\ln(1-\sigma(X))$$

This equivalence establishes that as a form approaches absolute closure, it traverses an infinite geodesic distance, identifying the isotropic state as the universal ideal boundary.

5 Empirical Analysis: Sacred Architectural Archetypes

Sacred cubit-based forms represent a "clean" corpus for testing geometric attractors due to their rigorous dimensional ordering and rejection of arbitrary extension.

5.1 Comparative Corpus Results

5.2 Analysis of Morphological Clustering

The data clusters into three stable families. Notably, the Queen's Chamber (11:10:12) manifests as a "near-cubic threshold," scoring higher closure than the King's Chamber. This suggests that internal architectural organization is non-monotone with respect to elevation, separating spatial placement from internal isotopic reconciliation.

Object Name	Dimensions (Cubits)	σ_Q	τ	Fingerprint Class
Most Holy (Tabernacle)	10 : 10 : 10	1.0000	0.0000	T_{cube}
Temple Inner Sanctuary	20 : 20 : 20	1.0000	0.0000	T_{cube}
Queen's Chamber	11 : 10 : 12	0.9945	0.2000	Near-Cubic Threshold
Great Pyramid	440 : 440 : 280	0.9796	1.0000	T_{ascent}
Bronze Altar	5 : 5 : 3	0.9704	1.0000	T_{ascent}
Ark of the Covenant	2.5 : 1.5 : 1.5	0.9587	-0.3333	T_{carrier}
Holy Place	20 : 10 : 10	0.8889	-0.3333	T_{carrier}

6 Isotropic Convergence in Discrete Prime Constellations

The transition from architectural manifolds to number theory is bridged by the normalization of prime k-tuples to the simplex. Both systems occupy the same normalized closure space, governed by the discrete geometry of prime gaps.

6.1 Gap Vector Normalization and Regime Indexing

For a prime k-tuple (e.g., twins, triplets), we define the normalized gap state q. The Regime Index $R(k)$ is defined by the hyperbolic arclength:

$$R(k) = \lfloor \Lambda(k)/\pi \rfloor$$

The early exceptional compression observed at $k = 5$ represents a low-dimensional artifact where specific modular exclusions force high specificity. As k increases, the system settled into an asymptotic universal law.

6.2 The Asymptotic Regime Ladder

The progression of constellations follows an idealized ladder derived from the growth of the closure deficit:

$$k_m \approx 0.90148 \cdot e^{2m}$$

The first four primary regime markers, where k crosses into new bands of structural closure, are:

- Regime 1: $k \approx 5$
- Regime 2: $k \approx 50$
- Regime 3: $k \approx 364$
- Regime 4: $k \approx 2688$

7 Discussion: Structural Survival and Symmetry Renormalization

The convergence of architectural dimensions and prime gap distributions into identical regime structures is the signature of Structural Survival under Dimensional Increase.

As ambient dimension d increases, the "angle budget" for regular convex forms becomes severely overconstrained. By $d \geq 5$, perfect convex regularity collapses via Coxeter diagram constraints into only three universal families: the Simplex, the Hypercube, and the Cross-polytope. This process constitutes a symmetry renormalization. The "Families" observed in architecture (Cubic, Ascent, Carrier) and the "Regimes" of prime tuples are both manifestations of this collapse. The e^2 multiplicative scaling is the signature of this stabilization, indicating that as complexity increases, exceptional local symmetries fade, leaving only the dimensionally robust archetypes of closure.

8 Conclusion

We have established that the normalization of n-tuples to the simplex provides a unified backbone for structural analysis. By measuring the departure from isotropy via σ and subsequent complex/hyperbolic mapping, we demonstrate that disparate systems resolve into identical regime structures. The isotropic zero-point state serves as the universal attractor for all normalized geometric manifolds, proving that structural morphology is governed by the geodesic distance to absolute closure.

APPENDIX: THE BRIDGE DICTIONARY

The following operator stack converts raw geometric data across taxonomic layers:

- **Circular Representation:** $\mathcal{B}(X) = i(e^{i\pi\sigma(X)} + 1)$ — Maps closure to a phase-coded semicircular trajectory.
- **Projective/Tangent Operator:** $\mathcal{T}(X) = -\frac{\Re\mathcal{B}(X)}{\Im\mathcal{B}(X)} = \tan(\frac{\pi\sigma}{2})$ — Reduces circular flow to a boundary-sensitive projective slope.
- **Hyperbolic/Geodesic:** $\Lambda(X) = -\ln(1-\sigma(X))$ — Measures the geodesic distance from the anchor $2i$ toward the ideal point 0 in $\mathbb{H}$.

References

[1] L. Charles Allard. *Angular Manifold Routing: Sublinear Compute Reduction via Hopf-Base Sector Discretization.* Zenodo, 2026. DOI: 10.5281/zenodo.19243034. URL: `https://zenodo.org/records/19243034`